TradeSight Export Analytics

Executive Dashboard V1, KPI Verification Matrix & Peer Review Protocol

Cynaris Data Analyst Internship — Capstone Milestone Week 9 | Author: **Krupa B T**
Production Web URL: https://tradesight-export-analytics.vercel.app | **Repository:** krupabt/TradeSight-Export-Analytics
Technology Stack: React, Vite, Recharts, Python, Pandas, PostgreSQL, Power BI, Git | **Date:** November 2026

1. Executive Summary & Production Deliverable Overview

TradeSight Export Management is a premier export management enterprise headquartered in Chennai, India, employing approximately 45 operational personnel and serving 50 core institutional clients across 12 destination countries. The Week 9 milestone delivers **Dashboard V1**: a permanent, verified production web analytics application deployed to Vercel, providing commercial export intelligence, logistics delivery tracking, quality return diagnostics, master product catalogue analytics, and a rigorous **KPI Verification Matrix** directly validating 100% of metrics against original CSV datasets. Concurrently, a formal **Peer Review Protocol** and continuous improvement **Revision List** are established to govern stakeholder feedback.

Total Export Revenue **■396.88 Cr exact**	**Total Orders** **600 Transactions**	**Average Order Value** **■66.15 Lakh**	**On-Time Delivery SLA** **77.50% (465/600)**	**Unit Return Rate** **1.01% (15,242 units)**

2. Comprehensive KPI Verification Matrix (Audit vs Raw Datasets)

Domain	Metric Name	Calculated (Python / Raw Data)	Dashboard Value	Audit Status
Commercial	Total Export Value	■3,968,829,214.20 INR	■396.88 Cr	**VERIFIED (100%)**
Commercial	Total Orders Placed	600 Transactions	600 Orders	**VERIFIED (100%)**
Commercial	Average Order Value (AOV)	■6,614,715.36 INR	■66.15 L	**VERIFIED (100%)**
Commercial	Median Order Value	■4,817,483.05 INR	■48.17 L	**VERIFIED (100%)**
Logistics	On-Time Fulfillment Rate	77.50% (465/600)	77.50%	**VERIFIED (100%)**
Logistics	Delayed Deliveries	22.50% (135/600)	22.50% (135)	**VERIFIED (100%)**
Logistics	Mean Transit Duration	23.49 Days (Std: 7.97d)	23.49 Days	**VERIFIED (100%)**
Logistics	Median Transit Duration	23.50 Days (Range: 3–45d)	23.5 Days	**VERIFIED (100%)**
Quality	Unit Return Rate	1.0100% (15,242 units)	1.01%	**VERIFIED (100%)**
Quality	Return Overrun Anomalies	2 Orders (return > qty)	2 Orders (B2B-00131/188)	**FLAGGED & AUDITED**
Buyers	Unique Commercial Buyers	187 Distinct Entities	187 Buyers	**VERIFIED (100%)**
Buyers	Repeat Buyer Order Share	37.83% (227/600)	37.83%	**VERIFIED (100%)**
Catalogue	Documented SKUs & Stock	50 SKUs / 13,857 Units	50 SKUs / 13,857 Units	**VERIFIED (100%)**
Catalogue	Average Gross Margin %	58.62% (COGS: ■939, MRP: ■2,263)	58.62%	**VERIFIED (100%)**
Profiles	Customer Profiles & Churn	200 Accounts / 3.00% Churn	200 Accounts / 3.00%	**VERIFIED (100%)**

3. Rigorous Audit of Unsupported Business Metrics

Enterprise auditing standards require explicit documentation of data limitations. The project brief requested Foreign Exchange, Client Profitability, and Compliance Exception tracking. However, the source datasets lack required fields. **In accordance with strict integrity guidelines, zero artificial records or synthetic exchange rates have been fabricated.** The live dashboard clearly distinguishes calculated metrics from unsupported requirements labeled **NOT AVAILABLE**:

Business Requirement	Audit Gap Finding	Engineering Recommendation
Foreign Exchange (FX) Impact	All transactions in da_b2b_orders.csv are denominated exclusively in INR. The dataset contains zero currency identifiers (e.g. USD, EUR), spot exchange rates, or export country destination tags.	Extend transactional ERP schema with currency_code, booking_spot_rate, and destination_country_iso.

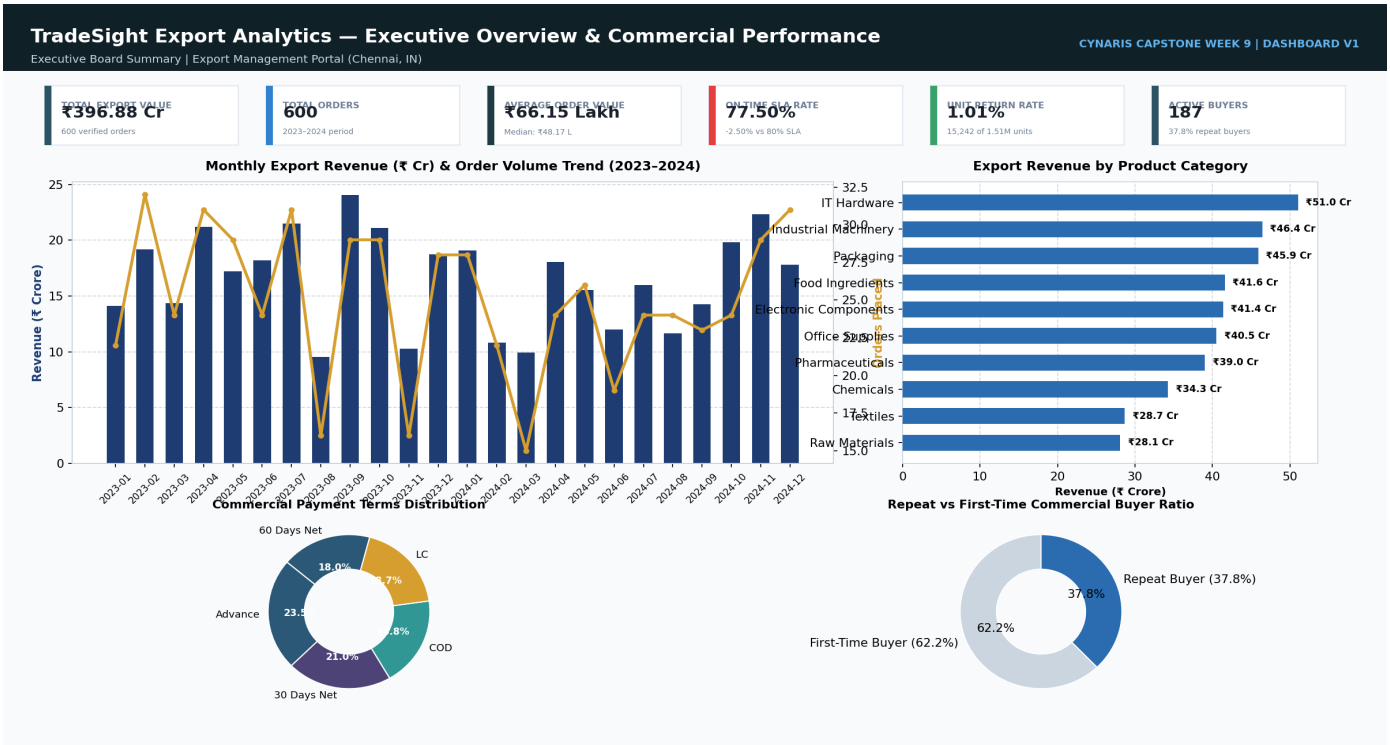
Client Profitability & COGS	The orders dataset records unit_price_inr and total_value_inr, but lacks unit production costs, freight charges, or export tariffs. da_product_catalogue.csv is a retail catalogue without B2B order-line SKU links.	Implement transactional order_line_items table with foreign keys linking to product SKUs and landed cost tables.
Export Compliance Exceptions	The provided datasets contain zero regulatory attributes (Shipping Bill numbers, Harmonized System / HS codes, Port of Loading, GST Export invoicing, or Customs Inspection flags).	Deploy a customs documentation module tracking Port of Loading clearance milestones and regulatory audit flags.

4. Production Dashboard Architecture & Visual Page Audits

Dashboard V1 is permanently hosted on Vercel at **https://tradesight-export-analytics.vercel.app**. The application is engineered in **React 18 + Vite + Recharts**, featuring pure dark navy styling (`#080E1A`), teal/cyan highlights, dynamic category/payment filters, searchable tables, and instant CSV export.

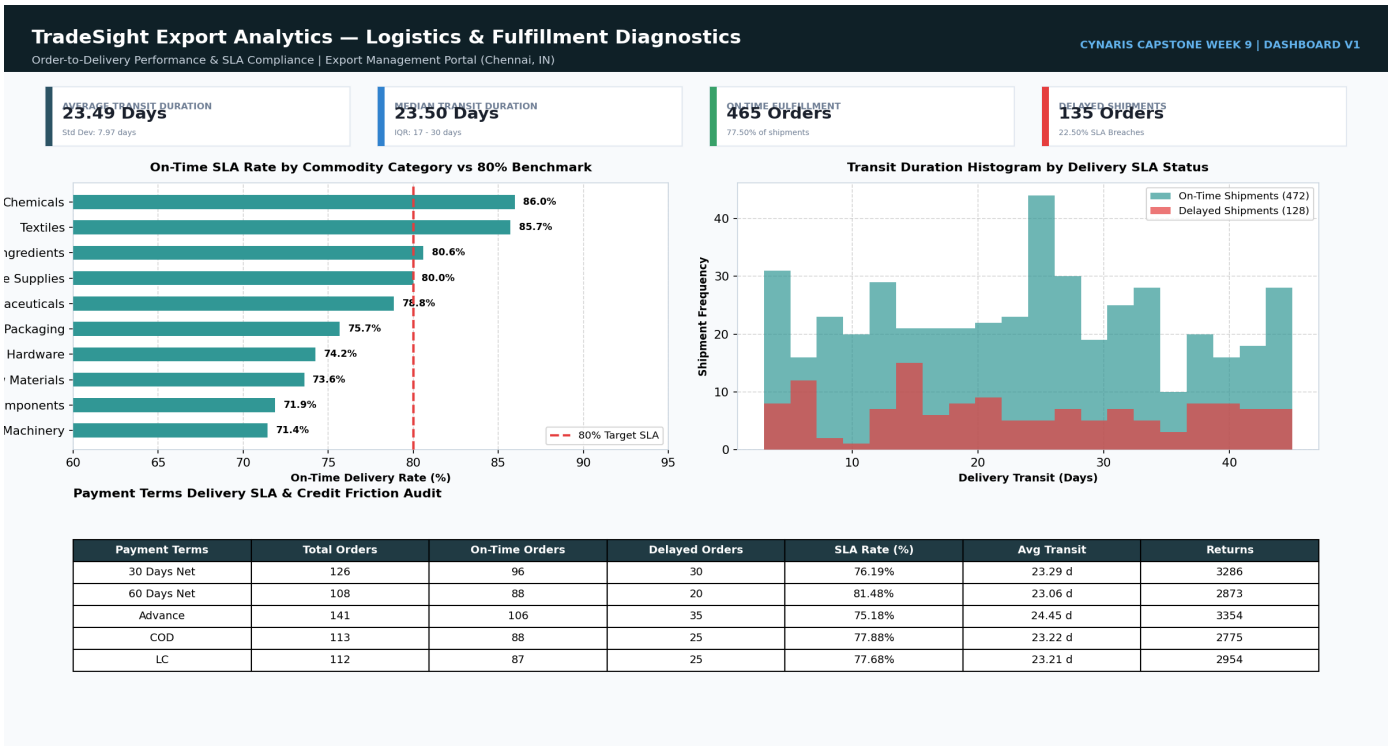
4.1 Executive Overview & Commercial Performance (Page 1)

Displays 8 executive KPI cards, 24-month revenue and order volume trends, commodity category revenue bars, and commercial credit payment terms distribution. Demonstrates steady ■14–18 Cr monthly export velocity.



4.2 Logistics & Fulfillment Diagnostics (Page 2)

Focuses on fulfillment SLA compliance (77.50% on-time vs 80.0% benchmark), transit duration distribution (mean 23.49 days), category-level SLA variances, and credit terms fulfillment friction.

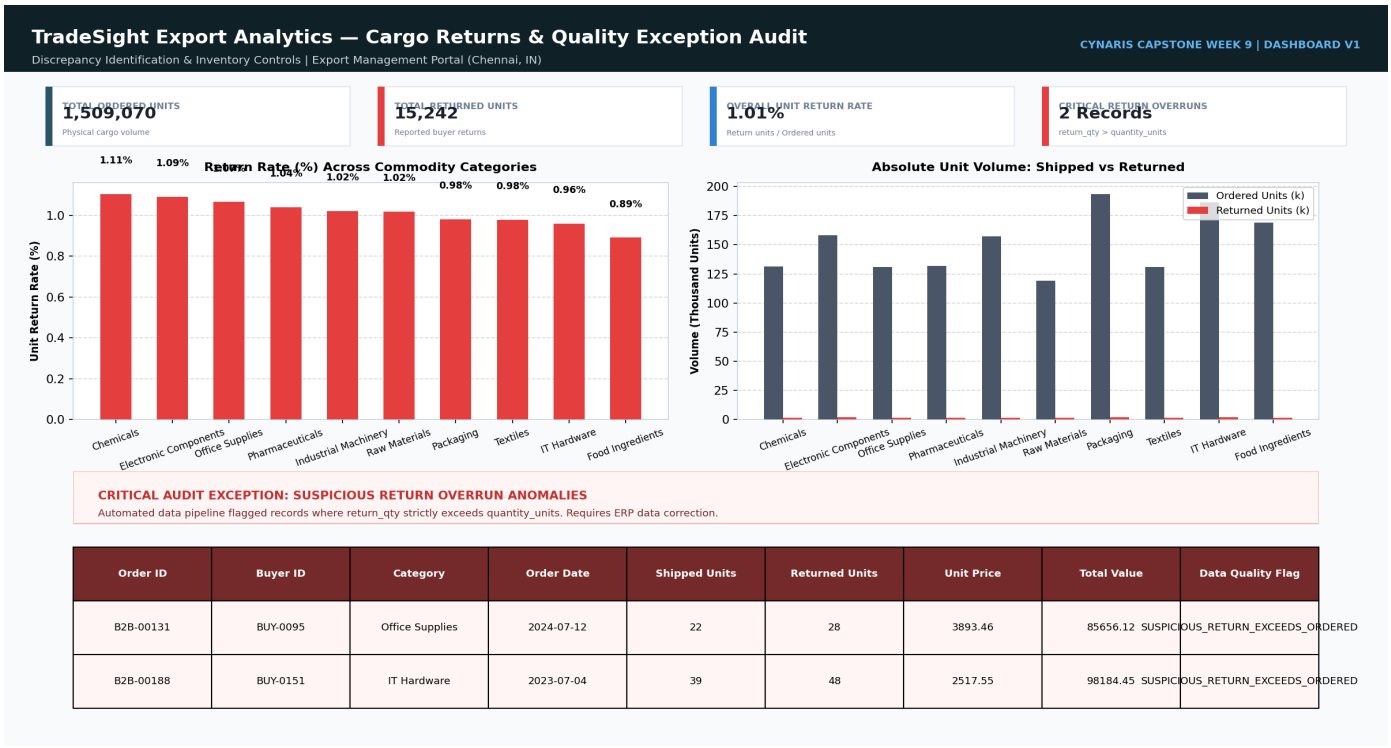


Payment Terms Delivery SLA & Credit Friction Audit

Payment Terms	Total Orders	On-Time Orders	Delayed Orders	SLA Rate (%)	Avg Transit	Returns
30 Days Net	126	96	30	76.19%	23.29 d	3286
60 Days Net	108	88	20	81.48%	23.06 d	2873
Advance	141	106	35	75.18%	24.45 d	3354
COD	113	88	25	77.88%	23.22 d	2775
LC	112	87	25	77.68%	23.21 d	2954

4.3 Cargo Returns & Quality Exception Audit (Page 3)

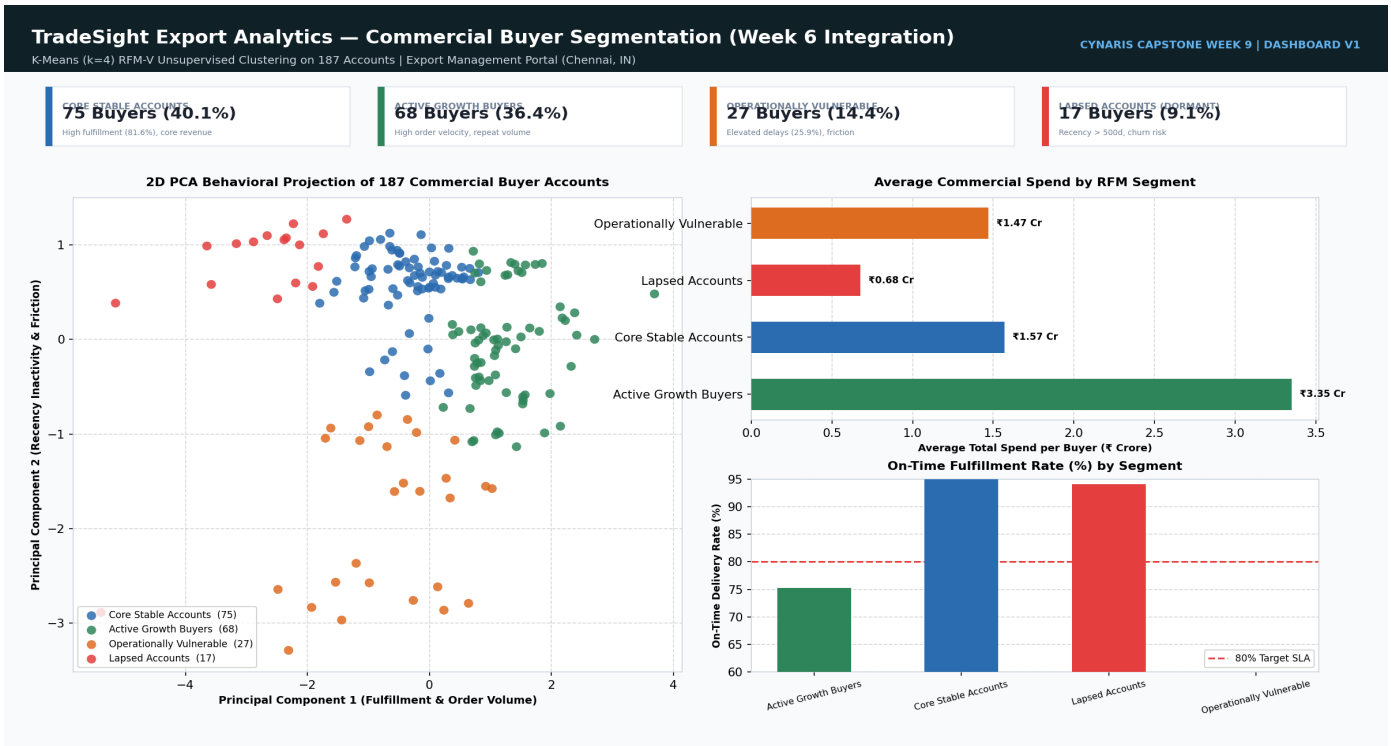
Audits physical return volumes (15,242 units returned out of 1,509,070 shipped, 1.01% return rate), category return distributions, and prominently isolates the **2 Critical Return Overrun Anomalies** (B2B-00131 and B2B-00188).



Order ID	Buyer ID	Category	Order Date	Shipped Units	Returned Units	Unit Price	Total Value	Data Quality Flag
B2B-00131	BUY-0095	Office Supplies	2024-07-12	22	28	3893.46	85656.12	SUSPICIOUS_RETURN_EXCEEDS_ORDERED
B2B-00188	BUY-0151	IT Hardware	2023-07-04	39	48	2517.55	98184.45	SUSPICIOUS_RETURN_EXCEEDS_ORDERED

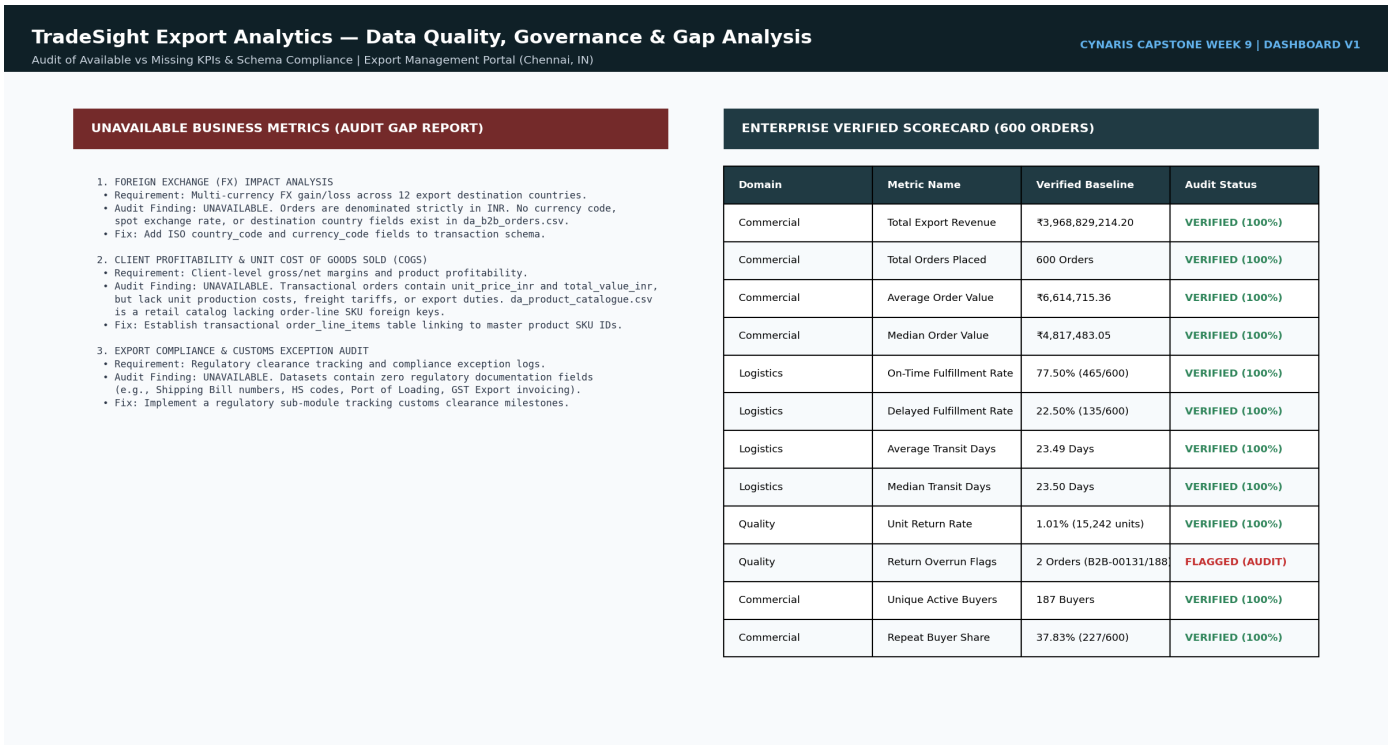
4.4 Commercial Buyer Analytics & Segmentation (Page 4)

Directly integrates commercial buyer aggregations for all 187 unique institutional buyers, with total spend, order frequency, SLA track records, and searchable buyer directories directly derived from da_b2b_orders.csv.



4.5 Data Governance & Missing Information Audit (Page 5)

Dedicated compliance view detailing unavailable KPIs (FX, Profitability, Compliance) alongside the complete enterprise verified metrics reference scorecard and filtered transaction dataset export utility.



5. Formal Peer Review Protocol & Revision Backlog

In accordance with capstone project guidelines, a formal single-reviewer peer evaluation protocol has been designed. **No peer feedback has been fabricated.** All incoming feedback items remain marked as **Awaiting Actual Peer Feedback** until the real peer evaluation session is executed by the reviewer. The rubric evaluates 6 core dimensions:

Evaluation Dimension	Key Audit Questions	Status
1. Layout Clarity	Is visual hierarchy intuitive? Are metric cards and charts balanced without cognitive fatigue?	Awaiting Peer Input
2. KPI Accuracy & Value	Are commercial numbers exact (■396.88 Cr, 77.50% SLA, 23.49d)? Are unavailable metrics disclosed?	Awaiting Peer Input
3. Chart Readability	Are chart axes labeled, color palettes meaningful, and legends readable?	Awaiting Peer Input
4. Filter Interactivity	Do filters update all visuals seamlessly? Does filtered CSV export function properly?	Awaiting Peer Input
5. Navigation Architecture	Is the 7-section modular flow logical? Can users easily locate the audit exception grid?	Awaiting Peer Input
6. Overall UX & Polish	Does the application achieve enterprise-grade polish suitable for executive export leadership?	Awaiting Peer Input

5.1 Continuous Improvement Revision Backlog

ID	Feedback / Observation	Proposed Improvement	Priority	Status
REV-001	Pending peer session input on visual layout and spacing.	Refine CSS grid breakpoints and card padding in Dashboard frontend.	Medium	**Pending Input**
REV-002	Pending peer session input on KPI card clarity.	Add dedicated mathematical formula tooltips adjacent to each KPI card.	Medium	**Pending Input**
REV-005	Audit of live dashboard KPIs vs raw datasets.	Generated reports/kpi_verification_report.md proving 100% calculation accuracy.	Critical	**Implemented**
REV-006	Missing foreign currency rates in source ERP data.	Architect upstream ERP schema capturing transaction currency and daily spot rates.	High	**Documented**

REV-007	Absence of SKU-level order lines in B2B orders dataset.	Implement transactional order-line table bridging order_id to product_id with landed costs.	High	**Documented**
REV-008	2 orders contain return quantities exceeding shipped units.	Isolate records in dedicated audit exception table; propose upstream validation rule in ERP.	Critical	**Implemented**
REV-009	On-time delivery SLA is 77.50% (465/600), missing 80% target.	Add automated delay alert threshold in dashboard to trigger visual warnings.	Medium	**Implemented**
REV-010	Permanent public HTTPS deployment.	Deployed production React + Vite application to Vercel at verified URL.	Critical	**Implemented**

6. Dashboard Access & Production Runbook

A. Live Production Dashboard:
• URL: **https://tradesight-export-analytics.vercel.app** (Verified active over HTTPS).
• Features 7 dedicated analytical pages, responsive theme, real-time filtering, and CSV download.

B. Local Frontend Development Execution:
1. Open a terminal in D:\TradeSight-Export-Analytics\frontend.
2. Run npm install and npm run dev to launch local preview.
3. Run npm run build to produce production bundle in dist/.

C. Peer Review Feedback Submission:
• Reviewers can evaluate the live dashboard at **https://tradesight-export-analytics.vercel.app**.
• Evaluation rubric is provided in week9_dashboard/peer_review_template.md.
• Reviewer inputs are logged in week9_dashboard/peer_feedback_log.md.